CMOR 504
Project Report

Project Report

Non-Isomorphic Local Complement Branch and Bound Scheme and its Applications to Graph Theory Fundamentals

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Abstract

Recent work in quantum computing focuses on understanding qubits in the lense of graph theory. One such operation includes the local complement. Despite its simple definiton, it can be become a redundant task if the graph is presumed to be dense and have local complements that are isomorphic to previous graphs. We're focused on constructing a recursive algorithm that finds all non-isomorphic graphs from the initial-state graph. Given the final listing of all the non-isomorphic graphs, we'll proceed to quantify the various properties of each graph including maximum stable set, maximum cliques, matchings, and perfect matchings.

Introduction

Graph states in [5] connect qubits and both their computation and communication. One such focus comes from [1] on the use of local complements as a transformation on the original stated graph. Despite the local complement operation being straightforward, finding all possible combinations of such local complements can be challenging when the node count is large or the graph is densely connected. In fact, it can be redundant to end up seeing graphs that are isomorphic to ones already seen from preivous iterations. This results in a difficulty in collecting the information of each individual graph. These include the following:

(i) *Maximum Clique*

(ii) *Maximum Stable/Independent Set*

(iii) *Matchings*

(iv) *Perfrect Matchings*

Hence, we propose a recursive algorithm that finds all non-isomorphic graphs to the original graph state. We'll proceed as follows:

(i) **Preliminary Material**: Reviewing the terminology and concepts to construct the recursive algorithm and the data to be collected on each graph.

(ii) **Methodology and Results**: Defining the recursive algorithm and presenting any results collected.

(iii) **Conclusions**: Summarize what's been done throughout this report.

(iv) **Limitations, Discussions, and Future Work**: Discuss the limitations and potential expansions to be performed on this report.

Preliminary Material

We start by defining the following terms.

Definition

A **graph G** is an ordered pair, defined as $G = (V, E)$, where V and E denote the vertices (or nodes) and edges, respectively. The neighborhood of a vertex $\mathbf{v} \in V$, denoted $N_G(\mathbf{v})$, is the set of vertices adjacent to $\mathbf{v}$.

We'll suppose for this report that G is simple and V and E are finite.

The following definitions will be referenced from [1, 2, 3, 4].

Definition

Let the graph $G = (V, E)$ be given.

(i) An **independent/stable set** is any collection of vertices $S \subseteq V$ such that no two vertices in S share an edge.

(ii) The **maximum independent/stable set** problem finds the largest independent set $S \subseteq V$ to the graph G.

(iii) A **clique** is a sest of vertices $K \subseteq V$ such that all pairs of vertiecs in K are adjacent to eachother.

(iv) The **maximum clique** problem finds the largest clique $K \subseteq V$ to the graph G.

(v) A **matching** is a set $M \subseteq E$ of pairwise disjoint edges.

(vi) A **perfect matching** is a matching that covers each node in G.

Let the graph $H = (V', E')$ be given. Additionally, let ψ_G and ψ_H denote the incidence functions that assigns edges to an unordered pair of vertices to both G and H, respectively.

The graph G is said to be **isomorphic** to H, denoted $G \cong H$, if there exists a bijective functions $f_V : V \to V'$ and $f_E : E \to E'$ such that

$$\psi_G(\mathbf{e}) = \mathbf{uv} \text{ if and only if } \psi_H(f_E(\mathbf{e})) = f_V(\mathbf{u})f_V(\mathbf{v}).$$

The **local complement** of a graph on a vertex $\mathbf{v} \in V$, denoted $G_{LC} = (V, E')$, is the complement of $N_G(\mathbf{v})$ where

$$E' \doteq (E \cup K_{N_G(\mathbf{v})}) \setminus (E \cap K_{N_G(\mathbf{v})}) = E \,\Delta\, K_{N_G(\mathbf{v})}.$$

See Figure 1 for reference.

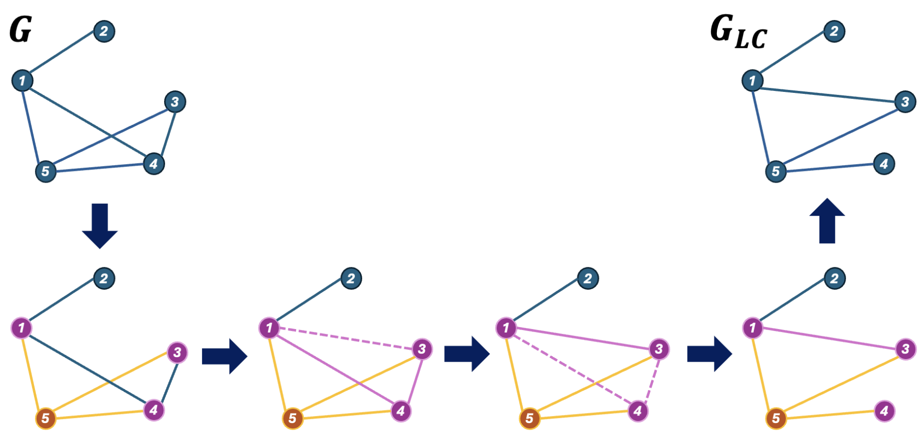

Figure 1: A guide on how the local complement of a graph yields the output graph.

Methodology

Our work on finding all combinations of local complements on a graph G will be done using a branch-and-bound scheme. This idea was inspired by [4] with the scheme being used to find the maximum stable/independent set and maximum cliques. We'll instead approach the problem by finding all possible combinations of such local complements (see Figure 2 for reference).

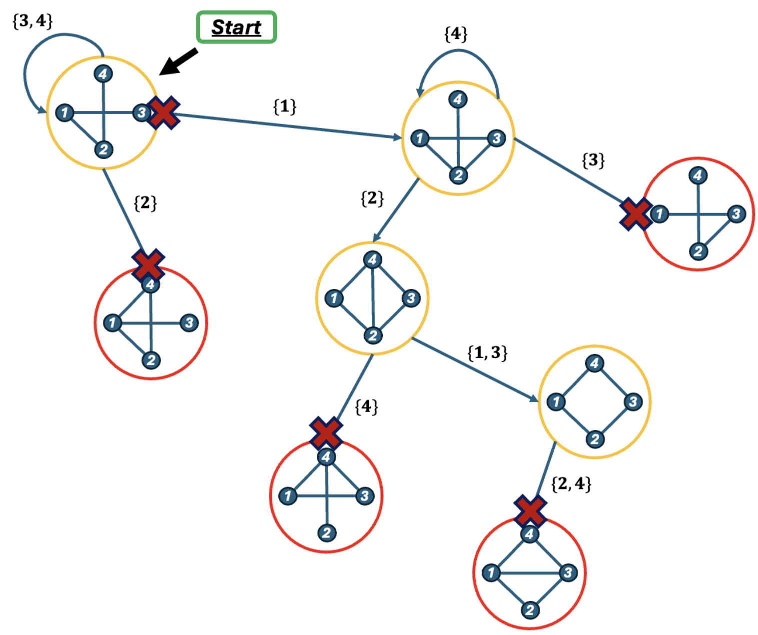

Figure 2: Branch-and-bound scheme overview. Note that the graph eventually returns to the initial graph state, though it may be isomorphic to a previously seen graph.

In constructing our algorithm, we define the function `isomorphic_status`(G) where G is a checked if it's isomorphic to the previous seen graphs in `graph_list`. Given this, we define the following function `gen_orbits(G, lvl)` where `lvl` is the recursion level index.

Algorithm 1 `gen_orbits`(G, `lvl`)

Require: Graph $G = (V, E)$, Recursion level index `lvl`
Ensure: Set of all non-isomorphic graphs reachable via local complements
  if `is_isomorphic`(G) **then**
      `return`
  end if
  `graph_list.append`(G)
  for $\mathbf{v} \in V(G)$ **do**
      if $|N_G(\mathbf{v})| \leq 1$ **then**
          `continue`
      end if
      $G_{\texttt{new}} =$ `gen_local_complement`($G, N_G(\mathbf{v})$)
      `gen_orbits`($G_{\texttt{new}}$, `lvl` $+ 1$)
  end for

Once all the local complements of the original graph G are collected, we continue to collect the following information on each graph:

(i) **Maximum Independent/Stable Set**

(ii) **Maximum Clique**

(iii) **Matchings**

(iv) **Perfect Matchings**

For these results, we use relaxed versions of their integer program formulations. Supposing the graph $G = (V, E)$ is given, we have the following linear programs:

Maximum Independent/Stable Set:

$$\begin{aligned} \max \quad & \sum_{\mathbf{v} \in V} x_{\mathbf{v}} \\ \text{s.t.} \quad & x_{\mathbf{u}} + x_{\mathbf{v}} \leq 1 \quad \forall\, \mathbf{uv} \in E \\ & x_{\mathbf{v}} \geq 0 \qquad\qquad \forall\, \mathbf{v} \in V \end{aligned}$$

Maximum Clique:

$$\begin{aligned} \max \quad & \sum_{\mathbf{v} \in V} x_{\mathbf{v}} \\ \text{s.t.} \quad & x_{\mathbf{u}} + x_{\mathbf{v}} \leq 1 \quad \forall\, \mathbf{uv} \notin E \\ & x_{\mathbf{v}} \geq 0 \qquad\qquad \forall\, \mathbf{v} \in V \end{aligned}$$

Matchings:

$$\begin{aligned} \max \quad & \sum_{\mathbf{e} \in E} x_{\mathbf{e}} \\ \text{s.t.} \quad & \sum_{\mathbf{e} \in \delta(\mathbf{v})} x_{\mathbf{e}} \leq 1 \quad \forall\, \mathbf{v} \in V \\ & x_{\mathbf{e}} \geq 0 \qquad\qquad \forall\, \mathbf{e} \in E \end{aligned}$$

Perfect Matchings:

$$\begin{aligned} \max \quad & \sum_{\mathbf{e} \in E} x_{\mathbf{e}} \\ \text{s.t.} \quad & \sum_{\mathbf{e} \in \delta(\mathbf{v})} x_{\mathbf{e}} = 1 \quad \forall\, \mathbf{v} \in V \\ & x_{\mathbf{e}} \geq 0 \qquad\qquad \forall\, \mathbf{e} \in E \end{aligned}$$

Note that in both the linear programs for both matchings and perfect matchings, we let $\delta(\mathbf{v})$ denote the set of all edges $\mathbf{e} \in E$ that have $\mathbf{v}$ is an endpoint.

Computational Results

Conclusions

Text on the conclusions material goes here . . . .

Limitations, Discussions, and Future Work

Text on the limitations, discussions, and future work goes here . . . .

Bibliography

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